

Filippov Sliding Mode Detection and Handling

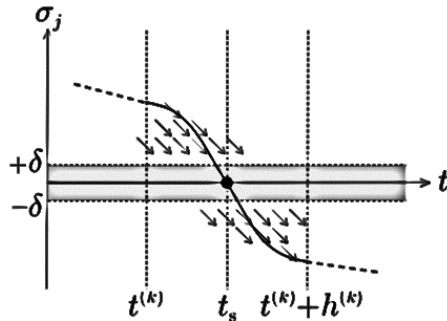
Switched ODEs

- ODE/IVP with switches: $\dot{x}(t) = f(t, x(t), p, \text{sign } \sigma(t, x(t), p))$ $x(t_0) = x_0(p)$
- 3 types of switches (the good, the bad, and the ugly):

Consistent switch

leaving $\sigma = 0$ in well-defined direction

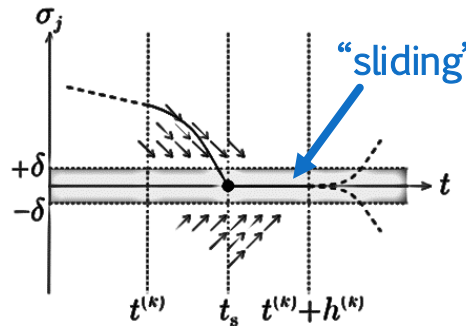
$$\frac{d\sigma^-}{dt_s}(t_s) \cdot \frac{d\sigma^+}{dt_s}(t_s) > 0$$



Inconsistent switch

staying at $\sigma = 0$

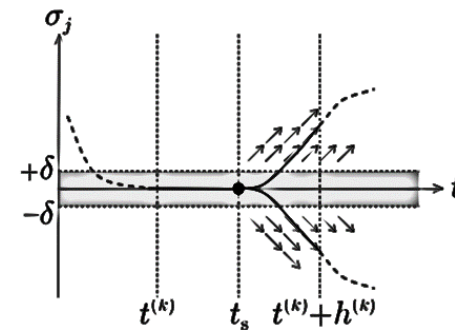
$$\frac{d\sigma^-}{dt_s}(t_s) < 0 \text{ and } \frac{d\sigma^+}{dt_s}(t_s) > 0$$



Bifurcation

leaving $\sigma = 0$ in both directions

$$\frac{d\sigma^-}{dt_s}(t_s) > 0 \text{ and } \frac{d\sigma^+}{dt_s}(t_s) < 0$$



Illustrations: C. Kirches, A Numerical Method for Nonlinear Robust Optimal Control with Implicit Discontinuities and an Application to Powertrain Oscillations, 2006

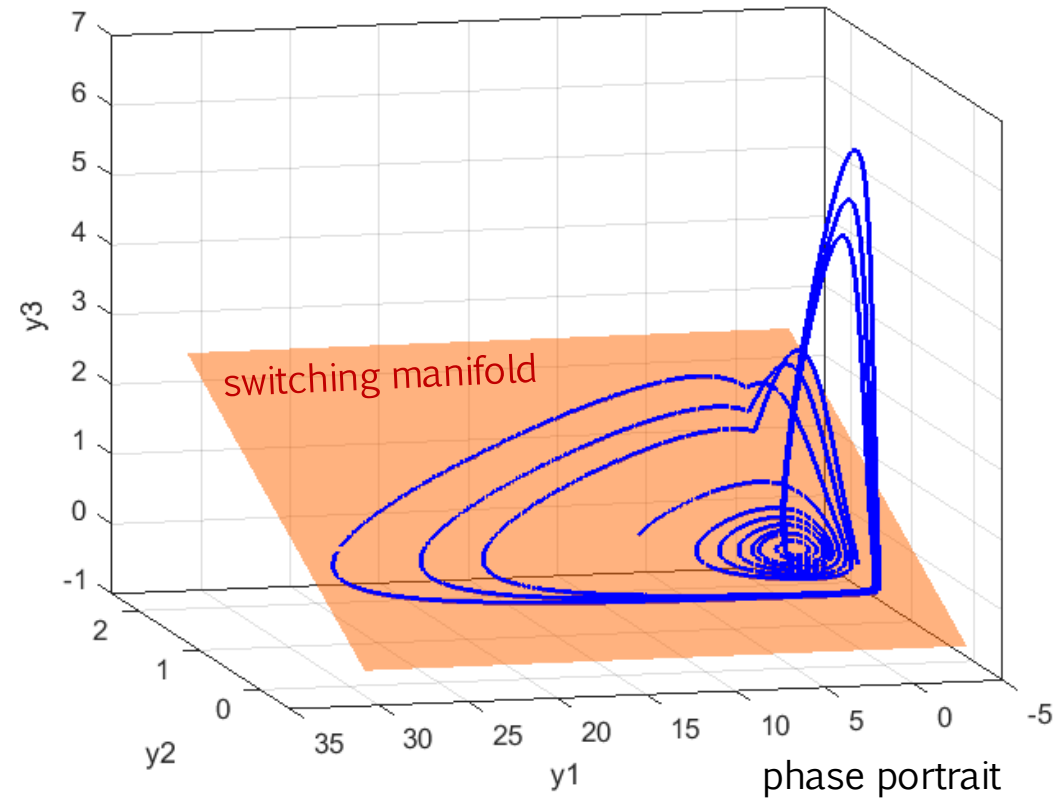
Example: Filippov-type 1 Predator – 2 Prey Model

■ 3D Predator-Prey Model [1], [2]

$$\dot{\mathbf{x}} = \begin{cases} \begin{pmatrix} (r_1 - \beta_1 P)p_1 \\ r_2 p_2 \\ (e q_1 \beta_1 p_1 - m)P \end{pmatrix} & \text{if } H(p_1, p_2, P) > 0, \\ \begin{pmatrix} r_1 p_1 \\ (r_2 - \beta_2 P)p_2 \\ (e q_2 \beta_2 p_2 - m)P \end{pmatrix} & \text{if } H(p_1, p_2, P) < 0, \end{cases}$$

where $\dot{\mathbf{x}} = (\dot{p}_1, \dot{p}_2, \dot{P})^T$, $(p_1, p_2, P) \in \mathbb{R}_{\geq 0}^3$ and

$$H(p_1, p_2, P) = \beta_1 p_1 - a_q \beta_2 p_2.$$



[1] Piltz, S.H., Porter, M.A., Maini, P.K.: Prey switching with a linear preference trade-off. SIAM J. Appl. Dyn. Syst. 13(2), 658–682 (2014)

[2] Carvalho, T., Duarte Novaes, D. & Gonçalves, L.F.: Sliding Shilnikov connection in Filippov-type predator-prey model. Nonlin Dyn 100, 2973–2987 (2020)

Filippov Sliding Mode – A Basic Example

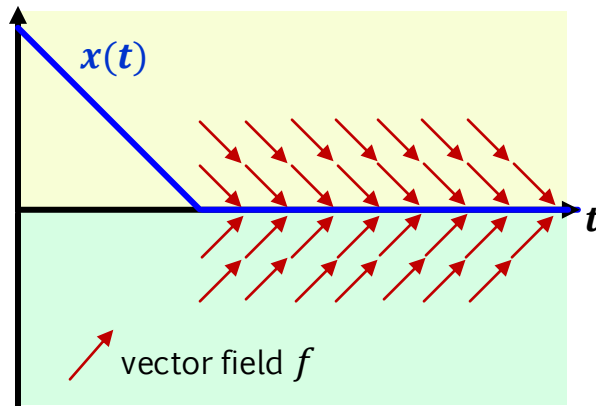
▪ Filippov-ODE: $\frac{dx}{dt} = \begin{cases} f_1(x(t)) & \text{if } \sigma(x) < 0 \\ f_2(x(t)) & \text{if } \sigma(x) > 0 \end{cases} \quad x(t_0) = x_0$

Switching manifold

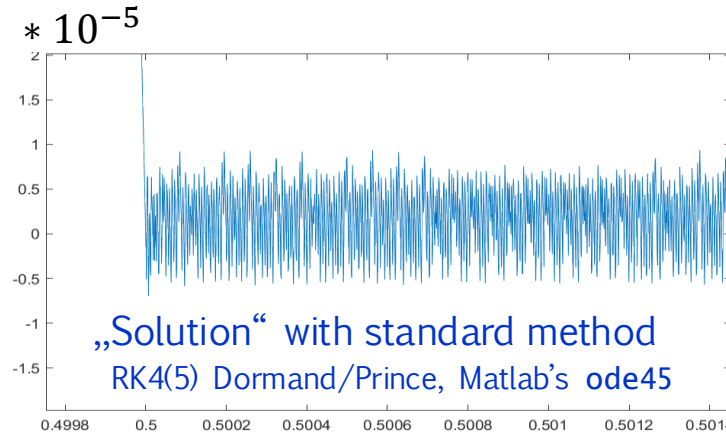
$$\Sigma := \{x \in \mathbb{R}^n \mid \sigma(x) = 0\}$$

▪ Example: $\frac{dx}{dt} = f(x) = \begin{cases} -1 & \text{if } x > 0 \\ +1 & \text{if } x < 0 \end{cases}$

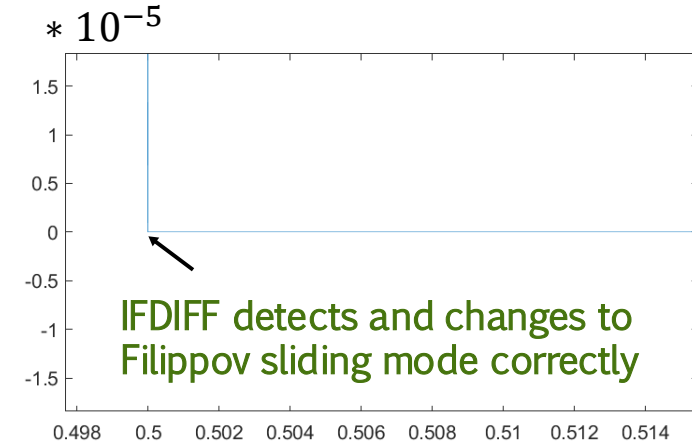
What about $x = 0$, i.e. Σ ?



Filippov sliding mode:
convex combination of models
 $f(t, x) = \alpha \cdot f_1(t, x) + (1 - \alpha) \cdot f_2(t, x)$

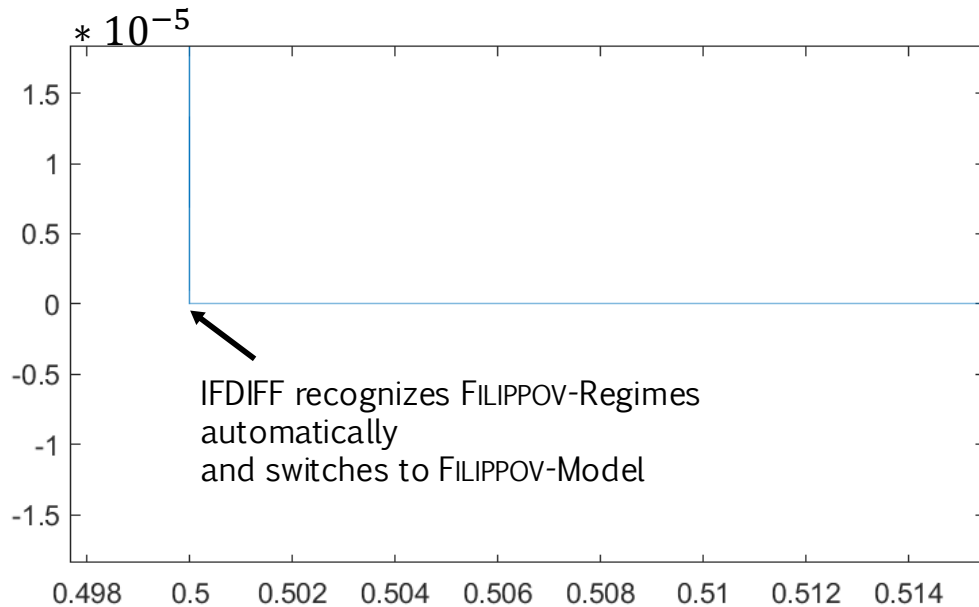


High-frequency oscillation (“model switching”) due to error control (tiny step sizes)
⇒ detected by IFDIFF, entering sliding mode

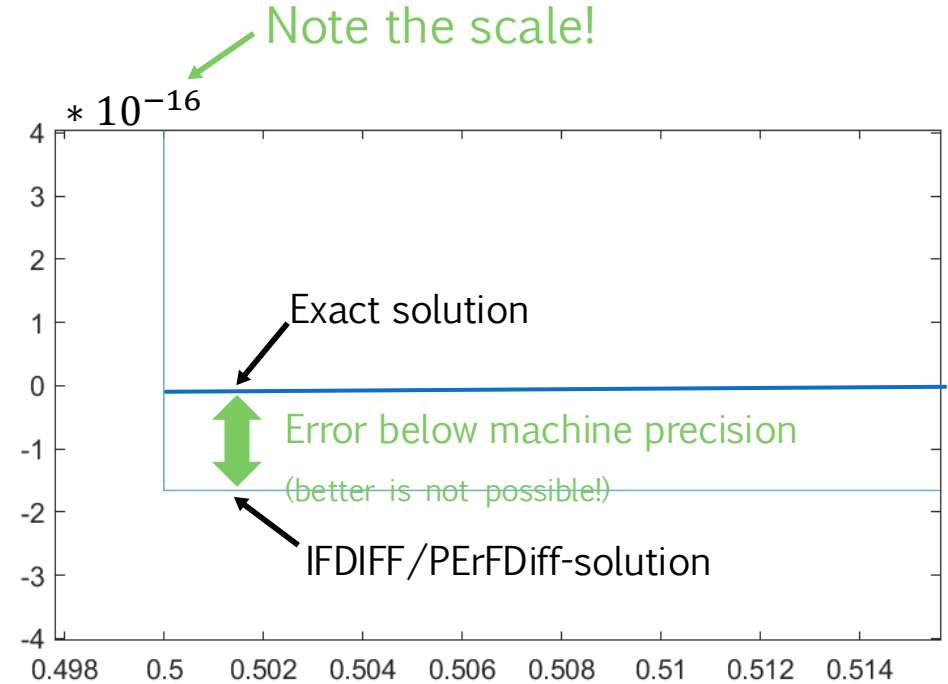


Automatic sliding mode detection by monitoring the switch frequency and calculating the convex parameter α

Filippov Sliding Mode – A Basic Example

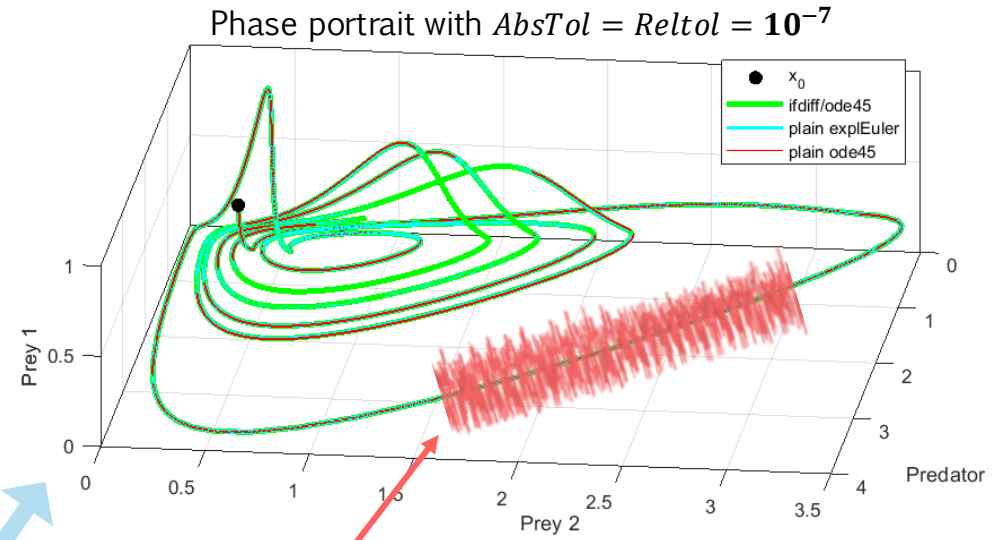
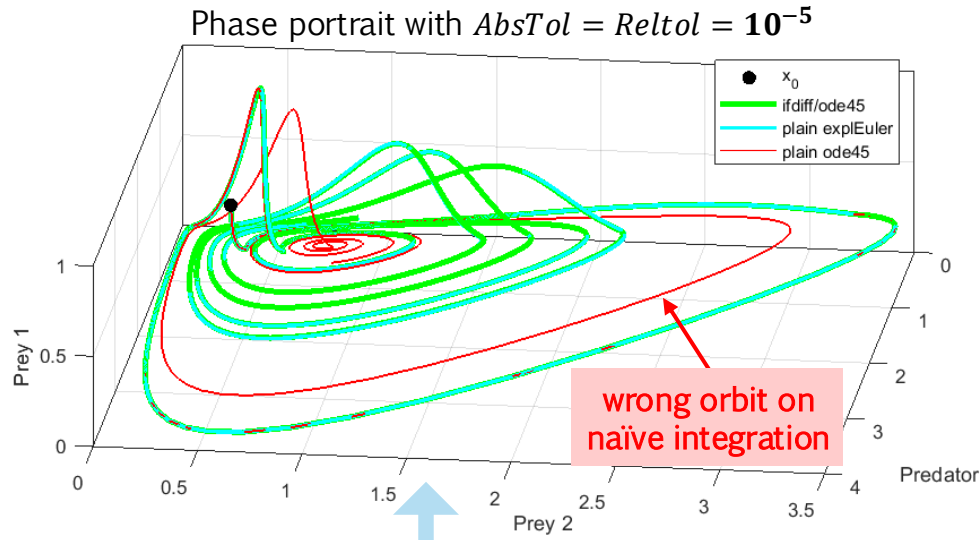


Correct solution with IFDIFF/PErFDiff



Comparison: True solution VS. IFDIFF solution

Some Results: Filippov-type 1 Predator – 2 Prey Model



RelTol AbsTol	10^{-5}		10^{-6}		10^{-7}		$10^{-8} 10^{-12}$	
	time	acc*	time	acc*	time	acc*	time	acc*
plain ode45	0.8s	✗	6.5s	✗	71.3s	⊙	>10min	?
IFDIFF/ode45	2.6s	✓	3.0s	✓	3.4s	✓	5.8s	✓

* compared to expl. Euler with step size 10^{-7}

? ode45 abort after ≈ 10 min on 8GB memory limit due to extreme oscillations on switching manifold

$\Rightarrow$ sensitivities inaccessible

$\Rightarrow$ IFDIFF gives consistent sensitivities

How to choose the convex combination?

Filippov: Computing the Convex-Combination

Consider the single-switched IVP

$$\frac{dx}{dt}(t) = f(t, x(t), p, \text{sgn}(\sigma(x))), \quad t \in T$$

$$x(t_0) = x_0$$

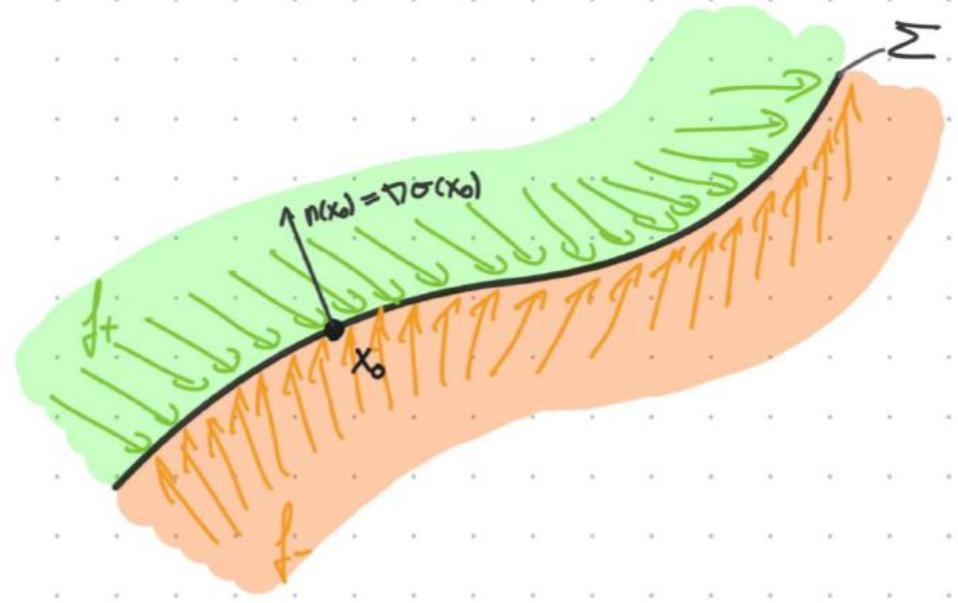
where we assume $\sigma(x_0) = 0$ and
inconsistent switching at x_0 .

Inconsistent switching:

$$n^T(x)f^-(x) > 0 \text{ and } n^T(x)f^+(x) < 0.$$

How to choose $\alpha(t, x)$ in

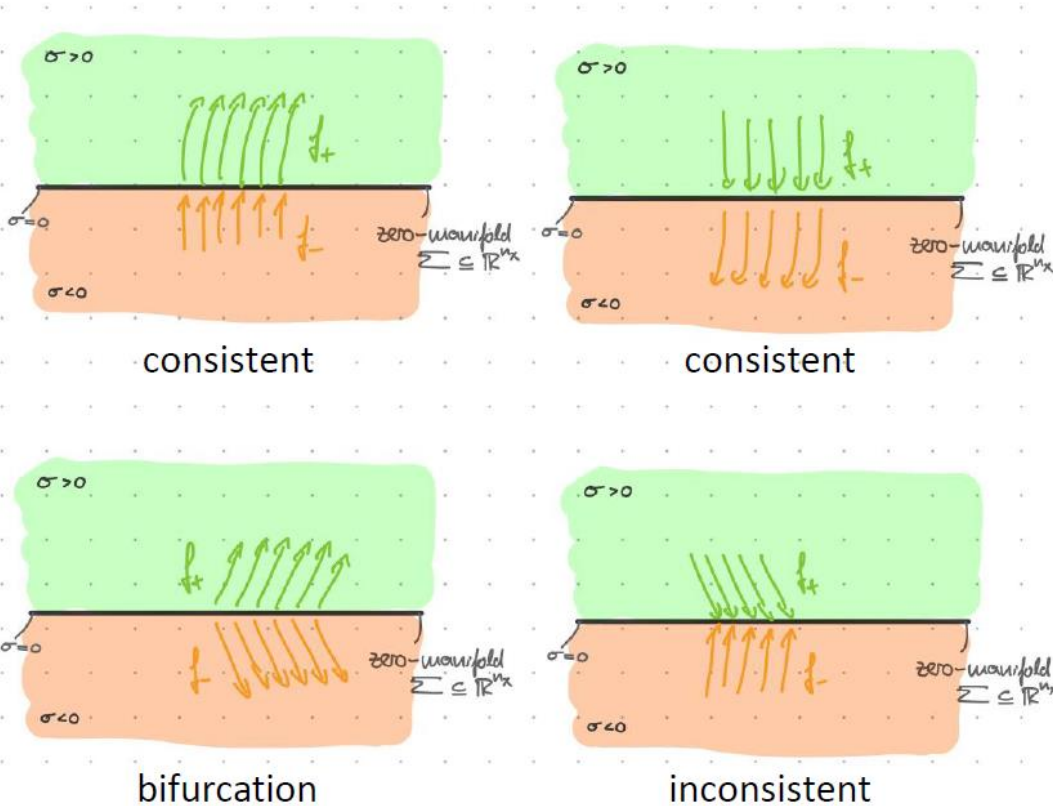
$$f_F(t, x) = \alpha(t, x)f_-(t, x) + (1 - \alpha(t, x))f_+(t, x) ?$$



Notation:

$$\Sigma = \{x \mid \sigma(x) = 0\}$$

Filippov: Computing the Convex-Combination



Not every root of σ comes with a well-defined solution!

For **inconsistent switching** no straightforward solution exists.

A **bifurcation** makes the problem ill-posed.

Let
$$n(x) = \frac{\nabla \sigma(x)}{\|\nabla \sigma(x)\|}.$$

The *transversality condition*

$$[n^T(x)f^-(x)] \cdot [n^T(x)f^+(x)] > 0$$

guarantees a well-defined model change.

Filippov: Computing the Convex-Combination

To continue the solution on the zero-manifold Σ :

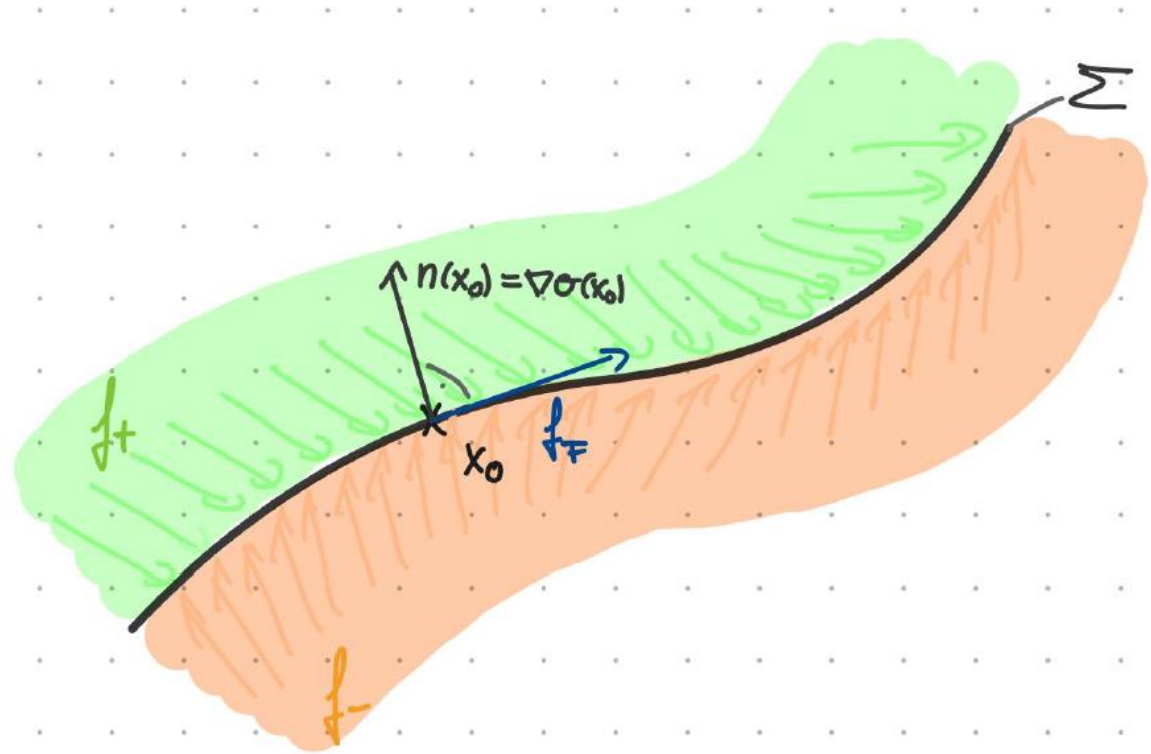
$f_F(t, x)$ must be orthogonal to the normal vector $n(x(t))$.

This is enough to compute $\alpha(t, x)$:

$$n^T(x) f_F(t, x) = 0$$

$\Rightarrow$

$$\alpha(t, x) = \frac{n^T(x) f_+(t, x)}{n^T(x) (f_+(t, x) - f_-(t, x))}$$





What does it cost?

Depends on the number of (actual) switches.
Factor 10-100 slower than “just integrating” in run-time.
(wrong results can always be computed fast)
Tremendous speed-up in model development.

Selected References

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- Tobias Reith. Algorithmic treatment of ordinary differential equations with implicit discontinuities in the right-hand side caused by non-differentiable conditional operators. Master's thesis, Ruprecht-Karls-Universität Heidelberg, 2019.
- Valentin von Trotha. Solving ordinary differential equations with implicit discontinuities. Master's thesis, University of Mannheim, 2021.
- Ilona Slutsker. Algorithmic Sensitivity Generation in Ordinary Differential Equation Systems with Implicit Discontinuities. Master's thesis, University of Mannheim, 2021.
- A. Sommer; H. G. Bock; E. Kostina. IFDIFF: A Matlab toolkit for ODEs with state-dependent switches (in preparation).